

# ACHINTYA RAI

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## EDUCATION

**University of California, San Diego**  
*Bachelor of Science in Computer Engineering*

Sept. 2024 – May 2027  
**GPA: 3.9/4.0**

- NSF REU Fellow, UC Scholars Research Fellow, Competitive Programming Club

## WORK EXPERIENCE

**Adobe**  
*Software Engineering Intern*

San Jose, CA  
June 2026 – Sept. 2026

- Analyzed **60.6M+** Databricks events via program analysis, surfacing **86,387** duplicates and a **47%** null rate for run-time monitoring.
- Detected a **26-hour** production outage via volume analysis across **1,272** segments, establishing the primary fault-detection benchmark.
- Designed a time-series anomaly detector to maintain code health across **10+** workflows and **30+** countries via statistical program analysis; built an MLOps pipeline for automated retraining and hourly inference.

**VALT Health**  
*Founding Software Engineer*

Remote  
Aug. 2025 – Feb. 2026

- Built GCP batch pipelines with fault tolerance via async processing, achieving **35%** speed increase and **60%** fewer redundant writes.
- Automated Linux setup via Bash scripts, cutting end-to-end testing overhead by **80%** for the engineering team.

**Decklar (formerly Roambee)**  
*Software Engineering Intern*

Sunnyvale, CA  
June 2024 – Aug. 2024

- Developed run-time monitoring modules for **Amazon**, **Coca-Cola**, and **Samsung**, optimizing Python pipelines for a **15%** capacity boost.
- Refactored analytics modules using advanced data structures, delivering **20%** faster reporting for **1M+** weekly events.

## RESEARCH

**Graph Theory & Neural Architecture**  
*Undergraduate Research Fellow (NSF REU)*

San Diego, CA  
March 2026 – Present

- Built a Python program analysis pipeline compressing **500GB** LLM checkpoints to **13GB** via weight-matrix extraction utilities.
- Optimized preprocessing scripts, achieving a **30x** runtime reduction (30 min → **<1 min**) on resource-constrained **autonomous systems** hardware.

## PROJECTS

**NanoGPT** | *C++, Python, PyTorch, Transformers, LLM Optimization* | [🔗](#)

Mar. 2026

- Compiled transformer sub-modules in **C++**, reducing per-token latency **25%** via modern C++ memory optimization.
- Engineered scalable dataloaders maintaining **100%** hardware utilization across multi-epoch training runs.

**SquaredUpChess** | *React, Django, AWS, Docker, WebSockets, Coolify, SQLite* | [🔗](#)

Aug. 2025 – Dec. 2025

- Established a **CI/CD pipeline** with automated end-to-end testing and error logging, ensuring **99.9%** code reliability across deployments.

## TECHNICAL SKILLS

**Languages:** Python, C, C++, TypeScript, SQL (Postgres/SQLite), Java, JavaScript, Bash, HTML/CSS

**Development:** Linux, build systems, static analysis, sanitizers, Docker, GitHub Actions, PyTest, Asyncio

**AI & Cloud:** PyTorch, AWS, GCP, Gemini/OpenAI API, LLMs, Asyncio, WebSockets, browser-use